

LOAN DEFAULT & CREDIT RISK ANALYSIS

Major Project – 3

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Project Objective

Analyze customer loan information and identify patterns associated with loan default. The project transforms financial data into meaningful credit-risk insights that can support risk monitoring and business decision-making.

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter/VS Code, Power BI Desktop, Power Query, DAX and GitHub.

1. Dataset Description

The cleaned dataset contains **255,347 loan records** and **22 columns**. It includes borrower demographics, income, credit score, employment, loan characteristics, interest rate, debt-to-income ratio, co-signer status and the Default indicator.

Main fields

LoanID, Age, Income, LoanAmount, CreditScore, MonthsEmployed, NumCreditLines, InterestRate, LoanTerm, DTIRatio, Education, EmploymentType, MaritalStatus, HasMortgage, HasDependents, LoanPurpose, HasCoSigner, Default and four derived risk-group fields.

2. Data Cleaning

The dataset was loaded with Pandas. Data types, statistics, missing values and duplicate records were checked. The final dataset contained no missing values and no duplicate rows. Four analytical groups were created: CreditScore_Group, DTI_Group, InterestRate_Group and LoanAmount_Group.

3. Descriptive Analysis

Metric	Result
Total Loans	255,347
Defaulted Loans	29,653
Non-Defaulted Loans	225,694
Default Rate	11.61%
Average Income	82,499.30
Average Loan Amount	127,578.87
Average Credit Score	574.26
Average Interest Rate	13.49%
Average DTI Ratio	0.50

4. Risk Segment Analysis

Interest rate and loan amount show the strongest visible gradients in observed default rates. Credit score and DTI also show smaller but consistent differences between risk groups.

Interest Rate Groups

InterestRate_Group	Loans	Defaults	Rate
Very High (20%+)	55,484	9,887	17.82%
High (16-20%)	43,969	6,250	14.21%
Medium (12-16%)	44,533	5,090	11.43%
Low (8-12%)	44,872	4,031	8.98%
Very Low (<8%)	66,489	4,395	6.61%

Loan Amount Groups

LoanAmount_Group	Loans	Defaults	Rate
High Loan Amount	63,837	9,961	15.60%
Medium-High Loan Amount	63,836	7,886	12.35%
Medium-Low Loan Amount	63,836	6,542	10.25%
Low Loan Amount	63,838	5,264	8.25%

Credit Score Groups

CreditScore_Group	Loans	Defaults	Rate
Poor (<580)	130,223	16,245	12.47%
Fair (580-669)	41,847	4,784	11.43%
Good (670-739)	32,073	3,409	10.63%
Very Good (740+)	51,204	5,215	10.18%

DTI Groups

DTI_Group	Loans	Defaults	Rate
Very High (70%+)	62,348	7,673	12.31%
High (50-70%)	63,781	7,646	11.99%
Medium (35-50%)	48,061	5,542	11.53%
Low (20-35%)	47,644	5,319	11.16%
Very Low (<20%)	33,513	3,473	10.36%

5.Categorical Analysis

Employment Type

EmploymentType	Loans	Defaults	Rate
Unemployed	63,824	8,650	13.55%
Part-time	64,161	7,677	11.97%
Self-employed	63,706	7,302	11.46%
Full-time	63,656	6,024	9.46%

Education

Education	Loans	Defaults	Rate
High School	63,903	8,230	12.88%
Bachelor's	64,366	7,789	12.10%
Master's	63,541	6,908	10.87%
PhD	63,537	6,726	10.59%

Co-Signer Status

HasCoSigner	Loans	Defaults	Rate
No	127,646	16,423	12.87%
Yes	127,701	13,230	10.36%

6. Python Visualizations

The Python analysis includes the required distribution charts, category-level default-rate charts, histograms, scatter plots, box plots and correlation heatmap. These visuals were used to identify and validate credit-risk patterns before dashboard development.

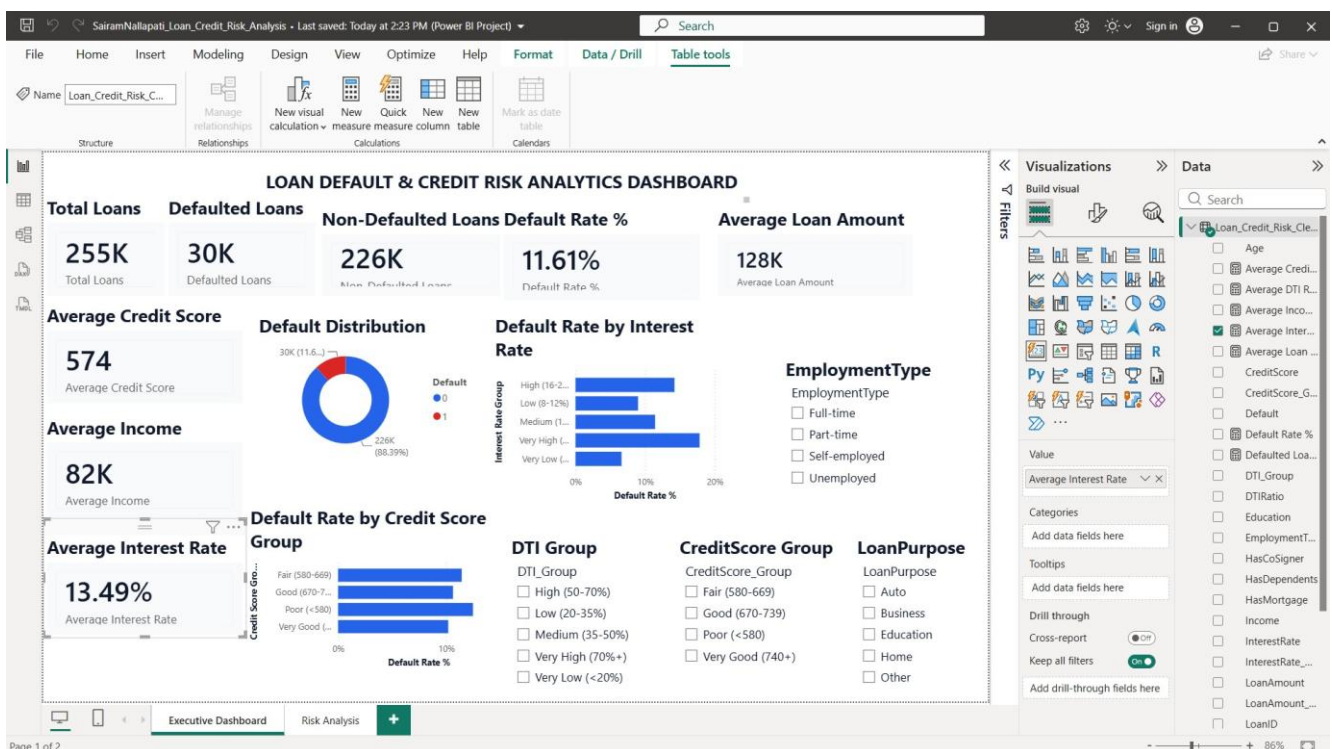
Key visualization set

- Loan Default Distribution — Pie Chart
- Loan Count / Default Rate by category — Bar Charts
- Credit Score, Loan Amount and Income — Histograms
- Credit Score vs Loan Amount — Scatter Plot
- Income vs Loan Amount — Scatter Plot
- Loan Amount / Credit Score / DTI by Default — Box Plots
- Default Rate by Income Group — Bar Chart
- Correlation Heatmap

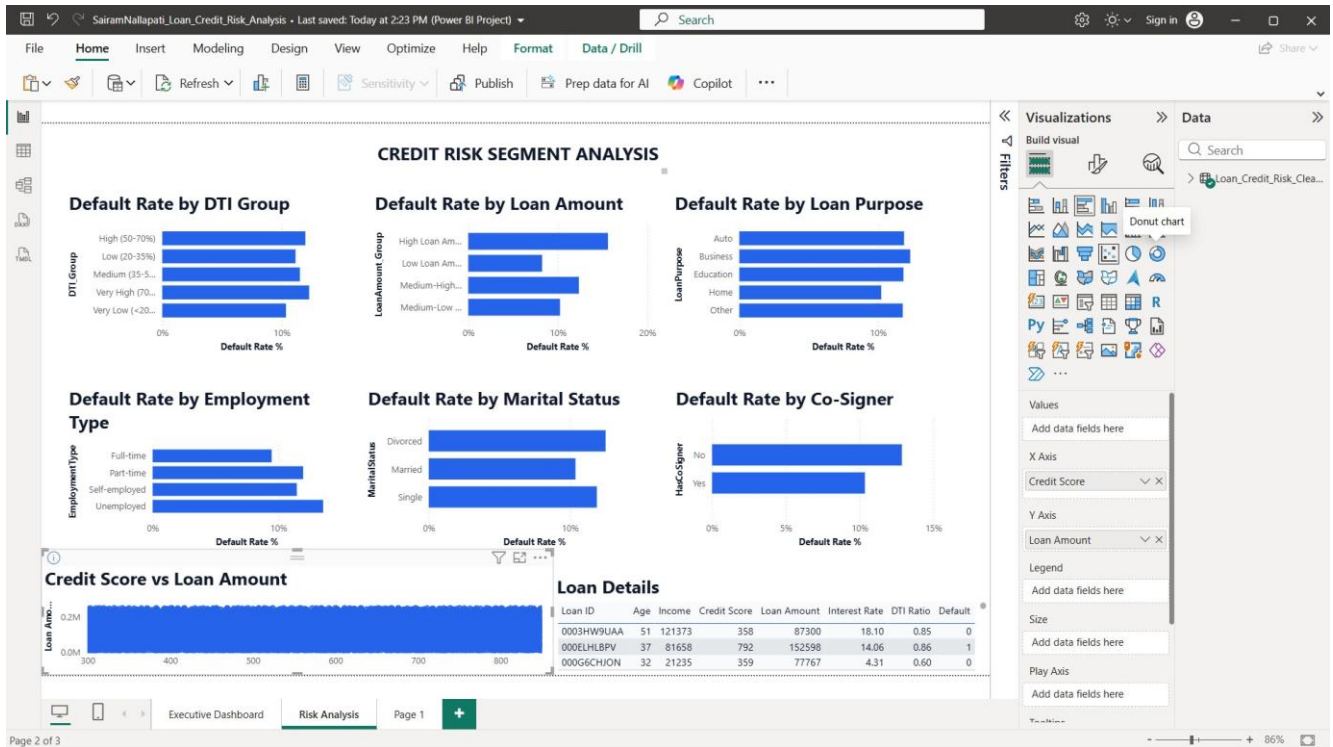
7. Power BI Dashboard

The dashboard is titled **Loan Default & Credit Risk Analytics Dashboard**. It contains KPI cards, default distribution, credit-score and interest-rate risk charts, slicers, DTI/loan-purpose/employment/marital/co-signer analysis, a Credit Score vs Loan Amount scatter plot and a loan-details table.

Executive Dashboard



Risk Analysis Dashboard



8. Power BI DAX Measures

Total Loans = COUNTROWS(Loan_Credit_Risk_Cleaned)

Defaulted Loans = CALCULATE(COUNTROWS(Loan_Credit_Risk_Cleaned), Loan_Credit_Risk_Cleaned[Default] = 1)

Non-Defaulted Loans = CALCULATE(COUNTROWS(Loan_Credit_Risk_Cleaned), Loan_Credit_Risk_Cleaned[Default] = 0)

Default Rate % = DIVIDE([Defaulted Loans], [Total Loans], 0)

Average Loan Amount = AVERAGE(Loan_Credit_Risk_Cleaned[LoanAmount])

Average Credit Score = AVERAGE(Loan_Credit_Risk_Cleaned[CreditScore])

Average Income = AVERAGE(Loan_Credit_Risk_Cleaned[Income])

Average Interest Rate = AVERAGE(Loan_Credit_Risk_Cleaned[InterestRate]) / 100

9. Key Business Insights

- Overall default rate is **11.61%**, with 29,653 defaults out of 255,347 loans.
- The Very High (20%+) interest-rate group has a default rate of **17.82%**, compared with **6.61%** for the Very Low (<8%) group.
- The High Loan Amount group has a default rate of **15.60%**, versus **8.25%** for the Low Loan Amount group.
- Poor credit scores (<580) have an observed default rate of **12.47%**, while Very Good scores (740+) are at **10.18%**.
- Default rate rises across the DTI bands, reaching **12.31%** in the highest-risk DTI group.
- The highest observed employment-segment default rate is **13.55%** for Unemployed customers.
- Customers without a co-signer show a higher observed default rate than customers with a co-signer.

10.Conclusion

Loan default risk is not uniform across the portfolio. Interest rate, loan amount, credit score, DTI and customer characteristics show measurable differences in observed default rates. The Power BI dashboard converts these findings into an interactive risk-monitoring view that can help identify higher-risk segments and support credit-risk review.

11.Project Deliverables

Loan_Credit_Risk_Cleaned.csv • Loan_Credit_Risk_Analysis.ipynb • SairamNallapati_Loan_Credit_Risk_Analysis.pbix • Loan_Credit_Risk_Project_Report.pdf • README.md